



HIPAA TECHNICAL SAFEGUARDS AUDIT

HIPAA Technical Safeguards Audit Report

Target

demo/patient-portal

Scan date

May 16, 2026 · 20:22 Venezuela Standard Time

Findings

17 total · Risk Score 0/100

Executive Summary

The Aegis scanner identified 17 potential HIPAA Technical Safeguards violations in the codebase (15 critical, 2 high). The aggregate risk score is 0/100. Critical and high findings should be remediated before any environment handling Protected Health Information is exposed to end users. Each finding below cites the controlling 45 CFR §164.312 provision and includes prescriptive remediation guidance.

Severity	Count
● CRITICAL	15
● HIGH	2
● MEDIUM	0
● LOW	0
Total	17

RISK SCORE

0 / 100

$\max(0, 100 - (10 \times \text{crit} + 5 \times \text{high} + 2 \times \text{med} + 1 \times \text{low}))$

Rule Coverage

- **AC-1** Access Control · 45 CFR §164.312(a)(1) · 2 findings
- **AC-2** Automatic Logoff · 45 CFR §164.312(a)(2)(iii) · 1 finding
- **AC-3** Person or Entity Authentication · 45 CFR §164.312(d) · 1 finding
- **AU-1** Audit Controls · 45 CFR §164.312(b) · 7 findings
- **EN-1** Encryption and Decryption · 45 CFR §164.312(a)(2)(iv) · 3 findings
- **IN-1** Integrity · 45 CFR §164.312(c)(1) · 1 finding
- **TR-1** Transmission Security · 45 CFR §164.312(e)(1) · 2 findings

Findings Index

ID	Rule	CFR	Severity	File:Line
d1d996c5	TR-1 Transmission Security	§164.312(e)(1)	● CRITICAL	config/database.js:16
95987445	EN-1 Encryption and Decryption	§164.312(a)(2)(iv)	● CRITICAL	models/Patient.js:22
1cd36610	EN-1 Encryption and Decryption	§164.312(a)(2)(iv)	● CRITICAL	models/Patient.js:31
2c5a2560	AU-1 Audit Controls	§164.312(b)	● CRITICAL	routes/admin.js:30
19b33753	AC-1 Access Control	§164.312(a)(1)	● CRITICAL	routes/appointments.js:9
3ff2e3d4	AU-1 Audit Controls	§164.312(b)	● CRITICAL	routes/appointments.js:9
991d5e6c	AC-3 Person or Entity Authentication	§164.312(d)	● CRITICAL	routes/auth.js:19
98927fd9	TR-1 Transmission Security	§164.312(e)(1)	● CRITICAL	routes/medications.js:10
69b1b039	AU-1 Audit Controls	§164.312(b)	● CRITICAL	routes/medications.js:12
6172ac9e	AU-1 Audit Controls	§164.312(b)	● CRITICAL	routes/medications.js:20
bc0a1aea	AU-1 Audit Controls	§164.312(b)	● CRITICAL	routes/medications.js:43
2bd25e2c	AU-1 Audit Controls	§164.312(b)	● CRITICAL	routes/medications.js:48
5376bb43	AC-1 Access Control	§164.312(a)(1)	● CRITICAL	routes/patients.js:24
0db68886	AU-1 Audit Controls	§164.312(b)	● CRITICAL	routes/patients.js:24
b9223866	EN-1 Encryption and Decryption	§164.312(a)(2)(iv)	● CRITICAL	routes/patients.js:41
b92c1223	AC-2 Automatic Logoff	§164.312(a)(2)(iii)	● HIGH	middleware/session.js:10
dffe8fa6	IN-1 Integrity	§164.312(c)(1)	● HIGH	routes/medications.js:50

Findings Detail

CRITICAL · 15 findings**45 CFR § 164.312(e)(1)****CRITICAL****Transmission Security** · TR-1 · ID d1d996c5

config/database.js:16-16

EVIDENCE

```
ssl: false,
```

WHY THIS IS A VIOLATION

Database connection has ssl explicitly disabled. PHI flowing to the database is transmitted in cleartext.

RECOMMENDED REMEDIATION

```
// 45 CFR § 164.312(e)(1) – recommended remediation
1. Enable SSL/TLS for database connections: set dialectOptions.ssl to true or 'require'
2. Use HTTPS URLs for all external API calls (https:// not http://)
3. Ensure all PHI transmission uses encrypted channels.
```

45 CFR § 164.312(a)(2)(iv)**CRITICAL****Encryption and Decryption** · EN-1 · ID 95987445

models/Patient.js:22-25

EVIDENCE

```
ssn: {
  type: DataTypes.STRING,
  allowNull: false,
},
```

WHY THIS IS A VIOLATION

PHI field 'ssn' is defined without encryption. Sensitive data should be encrypted at rest.

RECOMMENDED REMEDIATION

```
// 45 CFR § 164.312(a)(2)(iv) – recommended remediation
1. For Sequelize models: Use encryption hooks or encrypted field types for PHI fields like SSN.
2. For assignments: Encrypt PHI data before storing (e.g., patient.ssn = encrypt(req.body.ssn)).
3. Consider using libraries like crypto-js or sequelize-encrypted for field-level encryption.
```

45 CFR § 164.312(a)(2)(iv)**CRITICAL****Encryption and Decryption** · EN-1 · ID 1cd36610

models/Patient.js:31-33

EVIDENCE

```
diagnosis: {  
  type: DataTypes.TEXT,  
},
```

WHY THIS IS A VIOLATION

PHI field 'diagnosis' is defined without encryption. Sensitive data should be encrypted at rest.

RECOMMENDED REMEDIATION

```
// 45 CFR § 164.312(a)(2)(iv) – recommended remediation  
1. For Sequelize models: Use encryption hooks or encrypted field types for PHI fields like SSN.  
2. For assignments: Encrypt PHI data before storing (e.g., patient.ssn = encrypt(req.body.ssn)).  
3. Consider using libraries like crypto-js or sequelize-encrypted for field-level encryption.
```

45 CFR § 164.312(b)**CRITICAL****Audit Controls** · AU-1 · ID 2c5a2560

routes/admin.js:30-36

EVIDENCE

```
router.get('/audit', requireAuth, requireAdmin, async (req, res) => {  
  const rows = await AuditLog.findAll({  
    order: [['occurred_at', 'DESC']],  
    limit: 200,  
  });  
  res.json(rows);  
});
```

WHY THIS IS A VIOLATION

PHI route GET /audit does not call AuditLog.create(). Every read/write of PHI must produce an audit log entry.

RECOMMENDED REMEDIATION

```
// 45 CFR § 164.312(b) – recommended remediation  
Implement audit logging for all PHI access. If AuditLog model exists, ensure AuditLog.create() is called for all sensitive operations. Log: actor, action, resource type, resource ID, timestamp, and IP address.
```

45 CFR § 164.312(a)(1)**CRITICAL****Access Control** · AC-1 · ID 19b33753

routes/appointments.js:9-15

EVIDENCE

```
router.get('/:patientId', async (req, res) => {
  const rows = await Appointment.findAll({
    where: { patient_id: req.params.patientId },
    order: [['scheduled_for', 'DESC']],
  });
  res.json(rows);
});
```

WHY THIS IS A VIOLATION

Route GET /:patientId accesses PHI but has no authentication middleware. Found middleware: none

RECOMMENDED REMEDIATION

```
// 45 CFR § 164.312(a)(1) – recommended remediation
Add authentication middleware (e.g., requireAuth, authenticate, isAuthenticated) to all
routes that access Protected Health Information (PHI). Example:
router.get('/patients/:id', requireAuth, async (req, res) => { ... })
```

45 CFR § 164.312(b)**CRITICAL****Audit Controls** · AU-1 · ID 3ff2e3d4

routes/appointments.js:9-15

EVIDENCE

```
router.get('/:patientId', async (req, res) => {
  const rows = await Appointment.findAll({
    where: { patient_id: req.params.patientId },
    order: [['scheduled_for', 'DESC']],
  });
  res.json(rows);
});
```

WHY THIS IS A VIOLATION

PHI route GET /:patientId does not call AuditLog.create(). Every read/write of PHI must produce an audit log entry.

RECOMMENDED REMEDIATION

```
// 45 CFR § 164.312(b) – recommended remediation
Implement audit logging for all PHI access. If AuditLog model exists, ensure
AuditLog.create() is called for all sensitive operations. Log: actor, action, resource
type, resource ID, timestamp, and IP address.
```

45 CFR § 164.312(d)**CRITICAL****Person or Entity Authentication** · AC-3 · ID 991d5e6c

routes/auth.js:19-19

EVIDENCE

```
if (username === 'admin' && password === 'admin') {
```

WHY THIS IS A VIOLATION

Credential identifier compared against a string literal — hardcoded credentials bypass real authentication and are easily extracted from source.

RECOMMENDED REMEDIATION

```
// 45 CFR § 164.312(d) – recommended remediation
Remove hardcoded credentials. Use environment variables or secure credential management
systems. Implement proper authentication with password hashing (e.g., bcrypt) and secure
credential storage.
```

45 CFR § 164.312(e)(1)**CRITICAL****Transmission Security** · TR-1 · ID 98927fd9

routes/medications.js:10-10

EVIDENCE

```
|| 'http://internal-pharmacy-api/v1/prescriptions';
```

WHY THIS IS A VIOLATION

Insecure http:// URL ('http://internal-pharmacy-api/v1/prescriptions') used in HTTP client code. PHI in flight must travel over TLS — use https:// instead.

RECOMMENDED REMEDIATION

```
// 45 CFR § 164.312(e)(1) – recommended remediation
1. Enable SSL/TLS for database connections: set dialectOptions.ssl to true or 'require'
2. Use HTTPS URLs for all external API calls (https:// not http://)
3. Ensure all PHI transmission uses encrypted channels.
```


45 CFR § 164.312(b)**CRITICAL****Audit Controls** · AU-1 · ID 69b1b039

routes/medications.js:12-18

EVIDENCE

```
router.get('/:patientId', requireAuth, async (req, res) => {
  const rows = await Medication.findAll({
    where: { patient_id: req.params.patientId },
    order: [['started_at', 'DESC']],
  });
  res.json(rows);
});
```

WHY THIS IS A VIOLATION

PHI route GET `/:patientId` does not call `AuditLog.create()`. Every read/write of PHI must produce an audit log entry.

RECOMMENDED REMEDIATION

```
// 45 CFR § 164.312(b) – recommended remediation
Implement audit logging for all PHI access. If AuditLog model exists, ensure
AuditLog.create() is called for all sensitive operations. Log: actor, action, resource
type, resource ID, timestamp, and IP address.
```

45 CFR § 164.312(b)**CRITICAL****Audit Controls** · AU-1 · ID 6172ac9e

routes/medications.js:20-41

EVIDENCE

```
router.post('/', requireAuth, async (req, res) => {
  const med = await Medication.create({
    patient_id: req.body.patient_id,
    name: req.body.name,
    dosage: req.body.dosage,
    frequency: req.body.frequency,
    prescriber: req.body.prescriber,
  });

  try {
    await axios.post(PHARMACY_API, {
      patient_id: med.patient_id,
      drug: med.name,
      ... (9 more lines)
    });
  }
});
```

WHY THIS IS A VIOLATION

PHI route POST / does not call AuditLog.create(). Every read/write of PHI must produce an audit log entry.

RECOMMENDED REMEDIATION

```
// 45 CFR § 164.312(b) – recommended remediation
Implement audit logging for all PHI access. If AuditLog model exists, ensure
AuditLog.create() is called for all sensitive operations. Log: actor, action, resource
type, resource ID, timestamp, and IP address.
```

45 CFR § 164.312(b)**CRITICAL****Audit Controls** · AU-1 · ID bc0a1aea

routes/medications.js:43-46

EVIDENCE

```
router.get('/lookup/:rxcuri', requireAuth, async (req, res) => {
  const resp = await axios.get(`${PHARMACY_API}/lookup/${req.params.rxcuri}`);
  res.json(resp.data);
});
```

WHY THIS IS A VIOLATION

PHI route GET /lookup/:rxcuri does not call AuditLog.create(). Every read/write of PHI must produce an audit log entry.

RECOMMENDED REMEDIATION

```
// 45 CFR § 164.312(b) – recommended remediation
Implement audit logging for all PHI access. If AuditLog model exists, ensure
AuditLog.create() is called for all sensitive operations. Log: actor, action, resource
type, resource ID, timestamp, and IP address.
```

45 CFR § 164.312(b)**CRITICAL****Audit Controls** · AU-1 · ID 2bd25e2c

routes/medications.js:48-55

EVIDENCE

```
router.post('/discontinue', requireAuth, async (req, res) => {
  const { patient_id, status } = req.body;
  const [count] = await Medication.update(
    { status },
    { where: { patient_id } }
  );
  res.json({ updated: count });
});
```

WHY THIS IS A VIOLATION

PHI route POST /discontinue does not call AuditLog.create(). Every read/write of PHI must produce an audit log entry.

RECOMMENDED REMEDIATION

```
// 45 CFR § 164.312(b) – recommended remediation
Implement audit logging for all PHI access. If AuditLog model exists, ensure
AuditLog.create() is called for all sensitive operations. Log: actor, action, resource
type, resource ID, timestamp, and IP address.
```

45 CFR § 164.312(a)(1)**CRITICAL****Access Control** · AC-1 · ID 5376bb43

routes/patients.js:24-28

EVIDENCE

```
router.get('/:id', async (req, res) => {
  const patient = await Patient.findPk(req.params.id);
  if (!patient) return res.status(404).json({ error: 'not found' });
  res.json(patient);
});
```

WHY THIS IS A VIOLATION

Route GET /:id accesses PHI but has no authentication middleware. Found middleware: none

RECOMMENDED REMEDIATION

```
// 45 CFR § 164.312(a)(1) – recommended remediation
Add authentication middleware (e.g., requireAuth, authenticate, isAuthenticated) to all
routes that access Protected Health Information (PHI). Example:
router.get('/patients/:id', requireAuth, async (req, res) => { ... })
```

45 CFR § 164.312(b)**CRITICAL****Audit Controls** · AU-1 · ID 0db68886

routes/patients.js:24-28

EVIDENCE

```
router.get('/:id', async (req, res) => {
  const patient = await Patient.findByPk(req.params.id);
  if (!patient) return res.status(404).json({ error: 'not found' });
  res.json(patient);
});
```

WHY THIS IS A VIOLATION

PHI route GET /:id does not call AuditLog.create(). Every read/write of PHI must produce an audit log entry.

RECOMMENDED REMEDIATION

```
// 45 CFR § 164.312(b) – recommended remediation
Implement audit logging for all PHI access. If AuditLog model exists, ensure
AuditLog.create() is called for all sensitive operations. Log: actor, action, resource
type, resource ID, timestamp, and IP address.
```

45 CFR § 164.312(a)(2)(iv)**CRITICAL****Encryption and Decryption** · EN-1 · ID b9223866

routes/patients.js:41-41

EVIDENCE

```
patient.ssn = req.body.ssn;
```

WHY THIS IS A VIOLATION

PHI assignment 'patient.ssn' stores plaintext data from request. Encrypt sensitive data before storage.

RECOMMENDED REMEDIATION

```
// 45 CFR § 164.312(a)(2)(iv) – recommended remediation
1. For Sequelize models: Use encryption hooks or encrypted field types for PHI fields
   like SSN.
2. For assignments: Encrypt PHI data before storing (e.g., patient.ssn =
   encrypt(req.body.ssn)).
3. Consider using libraries like crypto-js or sequelize-encrypted for field-level
   encryption.
```

HIGH · 2 findings**45 CFR § 164.312(a)(2)(iii)****HIGH****Automatic Logoff** · AC-2 · ID b92c1223

middleware/session.js:10-10

EVIDENCE

```
maxAge: 86400000,
```

WHY THIS IS A VIOLATION

Session cookie maxAge is 86400000 ms (1440 minutes), exceeding the HIPAA-recommended 30-minute automatic logoff window.

RECOMMENDED REMEDIATION

```
// 45 CFR § 164.312(a)(2)(iii) – recommended remediation
Set session maxAge to 30 minutes (1800000 ms) or less. Example: session({ cookie: {
maxAge: 1800000 } })
```

45 CFR § 164.312(c)(1)**HIGH****Integrity** · IN-1 · ID dffe8fa6

routes/medications.js:50-53

EVIDENCE

```
const [count] = await Medication.update(
  { status },
  { where: { patient_id } }
);
```

WHY THIS IS A VIOLATION

Medication.update() is scoped by foreign key(s) ['patient_id'] rather than the primary key 'id'. This mass-updates every row sharing that parent and risks overwriting PHI for unrelated records.

RECOMMENDED REMEDIATION

```
// 45 CFR § 164.312(c)(1) – recommended remediation
Scope Model.update() to a specific primary key, not a foreign key. Example:
Model.update(data, { where: { id: specificId } }). A foreign-key where clause (e.g.
patient_id) will overwrite every row owned by that parent – a mass-update violation of
PHI integrity.
```

Appendix A · Rule Catalogue

AC-1 Access Control · 45 CFR § 164.312(a)(1)

CRITICAL

Implement technical policies and procedures for electronic information systems that maintain electronic protected health information to allow access only to those persons or software programs that have been granted access rights.

REMEDIATION GUIDANCE

Add authentication middleware (e.g., `requireAuth`, `authenticate`, `isAuthenticated`) to all routes that access Protected Health Information (PHI). Example: `router.get('/patients/:id', requireAuth, async (req, res) => { ... })`

AC-2 Automatic Logoff · 45 CFR § 164.312(a)(2)(iii)

HIGH

Implement electronic procedures that terminate an electronic session after a predetermined time of inactivity.

REMEDIATION GUIDANCE

Set session `maxAge` to 30 minutes (1800000 ms) or less. Example: `session({ cookie: { maxAge: 1800000 } })`

AC-3 Person or Entity Authentication · 45 CFR § 164.312(d)

CRITICAL

Implement procedures to verify that a person or entity seeking access to electronic protected health information is the one claimed.

REMEDIATION GUIDANCE

Remove hardcoded credentials. Use environment variables or secure credential management systems. Implement proper authentication with password hashing (e.g., `bcrypt`) and secure credential storage.

AU-1 Audit Controls · 45 CFR § 164.312(b)

CRITICAL

Implement hardware, software, and/or procedural mechanisms that record and examine activity in information systems that contain or use electronic protected health information.

REMEDIATION GUIDANCE

Implement audit logging for all PHI access. If `AuditLog` model exists, ensure `AuditLog.create()` is called for all sensitive operations. Log: actor, action, resource type, resource ID, timestamp, and IP address.

EN-1 Encryption and Decryption · 45 CFR § 164.312(a)(2)(iv)**CRITICAL**

Implement a mechanism to encrypt and decrypt electronic protected health information.

REMEDIATION GUIDANCE

1. For Sequelize models: Use encryption hooks or encrypted field types for PHI fields like SSN.
2. For assignments: Encrypt PHI data before storing (e.g., `patient.ssn = encrypt(req.body.ssn)`).
3. Consider using libraries like `crypto-js` or `sequelize-encrypted` for field-level encryption.

IN-1 Integrity · 45 CFR § 164.312(c)(1)**HIGH**

Implement policies and procedures to protect electronic protected health information from improper alteration or destruction.

REMEDIATION GUIDANCE

Scope `Model.update()` to a specific primary key, not a foreign key. Example: `Model.update(data, { where: { id: specificId } })`. A foreign-key where clause (e.g. `patient_id`) will overwrite every row owned by that parent — a mass-update violation of PHI integrity.

TR-1 Transmission Security · 45 CFR § 164.312(e)(1)**CRITICAL**

Implement technical security measures to guard against unauthorized access to electronic protected health information that is being transmitted over an electronic communications network.

REMEDIATION GUIDANCE

1. Enable SSL/TLS for database connections: set `dialectOptions.ssl` to `true` or `'require'`
2. Use HTTPS URLs for all external API calls (`https://` not `http://`)
3. Ensure all PHI transmission uses encrypted channels.

Appendix B · Scan Metadata

Scan target	demo/patient-portal
Generated at	2026-05-16 20:22:20 Venezuela Standard Time
Report ID	d0e76f7b-d328-43be-921e-2eda3e62a721
Findings	17
Risk score	0/100
Files with findings	8
Scanner version	0.1.0

Engine	Aegis Static Analyzer · tree-sitter + heuristics
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